IMAGES OF IRREDUCIBLE COMPONENTS UNDER SMOOTH MORPHISMS ARE IRREDUCIBLE COMPONENTS; SURJECTIVE CASE EXHAUSTS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 04XL: sources/stacks-chow/tags.txt, source lines 6399–6399. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : U \to \mathcal{X}$ be a smooth morphism from a scheme to a locally Noetherian algebraic stack. The closure of the image of any irreducible component of $|U|$ is an irreducible component of $|\mathcal{X}|$. If $U \to \mathcal{X}$ is surjective, then all irreducible components of $|\mathcal{X}|$ are obtained in this way.*

Proof. The map $|U| \to |\mathcal{X}|$ is continuous and open by Properties of Stacks, Lemma 04XL. Let $T \subset |U|$ be an irreducible component. Since U is locally Noetherian, we can find a nonempty affine open $W \subset U$ contained in T. Then $f(T) \subset |\mathcal{X}|$ is irreducible and contains the nonempty open subset $f(W)$. Thus the closure of $f(T)$ is irreducible and contains a nonempty open. It follows that this closure is an irreducible component.

Assume $U \to \mathcal{X}$ is surjective and let $Z \subset |\mathcal{X}|$ be an irreducible component. Choose a Noetherian open subset V of $|\mathcal{X}|$ meeting Z. After removing the other irreducible components from V we may assume that $V \subset Z$. Take an irreducible component of the nonempty open $f^{-1}(V) \subset |U|$ and let $T \subset |U|$ be its closure. This is an irreducible component of $|U|$ and the closure of $f(T)$ must agree with Z by our choice of T. □

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